David Vos

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Research interests

I study machine learning methods for search, personalized recommendations, and data management. My interests include text representation learning, in-context learning for ranking, and generative modeling for recommendations. My research at Bloomberg and in collaboration with DPG Media has addressed information retrieval in the news domain, where rapidly evolving content and short item lifetimes pose unique modeling challenges. In the long term, I aim to develop accessible information retrieval systems that account for their societal implications and serve diverse user information needs.

Publications

Prior-Data Fitted Networks as Tabular Foundation Models for Ranking in Low-Data Settings.
David Vos, Samarth Bhargav, Maarten de Rijke, Harrie Oosterhuis.
ACM Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR), 2026.

Reliable and Controllable Learned Sparse News Recommendations for Short-History Users.
David Vos, Andrew Yates, Maarten de Rijke.
ACM SIGIR Conference on Information Retrieval in the Asia Pacific (SIGIR-AP), 2026.

A Study of Normative Diversity Metrics in News Recommendations.
David Vos, Jin Huang, Maarten de Rijke.
ACM Transactions on Recommender Systems (TORS), 2026.

Towards Parameter-Efficient Automation of Data Wrangling Tasks with Prefix-Tuning.
David Vos, Till Döhmen, Sebastian Schelter.
NeurIPS 2022 First Table Representation Learning Workshop. **Best paper runner-up.**

Feature-based Propensities for Counterfactual Learning to Rank.
Stan Fris, **David Vos**, Harrie Oosterhuis.
ACM SIGIR Conference on Information Retrieval in the Asia Pacific (SIGIR-AP), 2026.

Efficient Optimization of Hierarchical Identifiers for Generative Recommendation.
Odysseas Boufalis, Polytimi Gkotsi, Joshua Rosenthal, Federica Valeau, **David Vos**.
European Conference on Information Retrieval (ECIR), 2026.

Revisiting Language Models in Neural News Recommender Systems.
Yuyue Zhao, Jin Huang, **David Vos**, Maarten de Rijke.
European Conference on Information Retrieval (ECIR), 2025.

Manuscript under review

Beyond Topical Relevance: Retrieval with Sorting Instructions over Latent Document Properties.
David Vos, Moniba Keymanesh, Aldo Porco, Kushagr Arora, Daniel Preoțiuc-Pietro, Hamed Zamani.

Experience

Research Intern, AI Group – News & Research Retrieval Jun. 2026 – Sep. 2026
Bloomberg, New York
- Investigated how retrieval models can follow sorting instructions over latent document properties, extending relevance beyond topical similarity. Resulting manuscript under review.
- Supervised by Moniba Keymanesh, Aldo Porco, Kushagr Arora, Daniel Preoțiuc-Pietro, and Hamed Zamani.

Visiting Researcher, Intelligent Data Engineering (INDE) Lab Aug. 2022 – Nov. 2022
University of Amsterdam
- Researched parameter-efficient adaptation of language models for data wrangling using prefix-tuning.
- Supervised by Sebastian Schelter and Marlies van der Wees.

Machine Learning Engineer Intern 2019 & 2021–2022
DPG Media – Recommendation & Search *Feb.–Jun. 2019; Nov. 2021–Jun. 2022*
- Conducted research in the Recommendation & Search team, including a BSc thesis on automating news article recommendations for a weekly newsletter.

Software & Data Engineering Apr. 2019 – Apr. 2023
Vaccinatieregister, Finly & A-INSIGHTS
- Developed production software and data pipelines for healthcare, customer support, and market intelligence.

Teaching Assistant Aug. 2017 – Nov. 2021
University of Amsterdam
- Led tutorials and graded assignments in BSc and MSc courses in mathematics, programming, and logic.

Education

PhD Candidate in Machine Learning & Information Retrieval Sep. 2023 – Present

Information Retrieval Lab – University of Amsterdam

- Research on responsible retrieval and ranking for news search and recommendations, in collaboration with DPG Media. Supervised by Maarten de Rijke and Harrie Oosterhuis.

MSc Artificial Intelligence (cum laude) Sep. 2019 – Sep. 2022

University of Amsterdam, GPA: 8.2/10

- Thesis: *Language control prefixes: Conditional prefix-tuning for efficient multilingual data-to-text generation in low-resource languages.*
- Supervisors: Sebastian Schelter and Marlies van der Wees.

BSc Artificial Intelligence (cum laude) Sep. 2016 – Jun. 2019

University of Amsterdam, GPA: 8.3/10

- Thesis: *Automating a weekly newsletter for news article recommendations at De Volkskrant.*
- Supervisors: Harrie Oosterhuis, Anne Schuth, and Lucas de Haas.
- Honours Programme: projects on clothing design generation with Siamese networks and GANs, and NLP-based sustainability benchmarking of multinational companies.

Erasmus Exchange Sep. 2018 – Dec. 2018

University of Edinburgh

- Elective courses in computer science and anthropology.

Academic activities

Talks & lectures

University of Amsterdam and external venues

- University MSc and BSc lectures on text representation learning, recommender systems, and transformers.
- Google NYC and Bloomberg ML&NLP Guild and Search Services: Retrieval with instructions beyond topical relevance.
- Office of the Dutch State Secretary of Digital Affairs: Prior-data assumptions for efficient AI development.
- Academic presentations at ECIR, NeurIPS workshops, and the Dutch Seminar on Data Systems Design.
- ING Data Science: Large language models for data management.

Teaching & supervision

University of Amsterdam

- Supervised master's theses on generative recommendations, feedback bias in recommender systems, gender bias introduced by fine-tuning large language models, feature-based propensity estimation for unbiased learning to rank, and long context convolutions for efficient tabular foundation models.
- Led tutorials and supervised students in MSc courses on Recommender Systems and Information Retrieval.

Service & organisation

IRLab Amsterdam

- Represent IRLab on the PhD council of the university's Institute of Informatics.
- Co-organised the European Summer School on Information Retrieval (ESSIR), 2025.
- Chair weekly research meetings and reading groups on neural information retrieval, recommender systems, and generative retrieval.

Training & honours

- Oxford Machine Learning School (OxML), 2025.
- European Summer School on Information Retrieval (ESSIR), 2025.
- The Next Web T500, 2018–2022: recognised among promising tech talent under 25.
- Cambridge English Certificate of Proficiency (C2).